THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The figure rotated by 72 degrees about its centre coincides with itself when its order of symmetry is: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 4

2. An isosceles trapezium has how many lines of symmetry? 1

(a) 0 **(b)** 2 **(c)** 4 **(d)** 1

3. The smallest angle through which a regular decagon must be rotated to coincide with itself is (degrees): 1

(a) 36 **(b)** 66 **(c)** 46 **(d)** 26

4. The mirror image of the letter H in a vertical mirror to its right looks like: 1

(a) an upside-down letter **(b)** a reversed letter

(c) the same letter **(d)** a different letter

5. A figure has rotational symmetry of order 5. After rotating through 144° it looks the same 2 time(s). The angle of one such turn is (degrees): 1

(a) 102 **(b)** 82 **(c)** 72 **(d)** 62

6. A regular polygon has 12 lines of symmetry. The number of its sides is: 1

(a) 12 **(b)** 11 **(c)** 13 **(d)** 14

7. Which of these has no symmetry at all? 1

(a) H **(b)** F **(c)** O **(d)** X

8. The figure rotated by 90 degrees about its centre coincides with itself when its order of symmetry is: 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5

9. The order of rotational symmetry of a rhombus (not a square) is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

10. The smallest angle through which a regular octagon must be rotated to coincide with itself is (degrees): 1

(a) 35 **(b)** 45 **(c)** 55 **(d)** 75

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the mirror image of the word WOW in a vertical mirror (the reflection reads reversed) and say if the word looks the same. 2

12. Draw (describe) all lines of symmetry of a kite. 2

13. Find the angle of rotational symmetry and order for a regular polygon of 12 sides. 2

14. Write the mirror image of the word HAT in a vertical mirror (the reflection reads reversed) and say if the word looks the same. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A regular polygon has rotational symmetry of angle 90°. How many sides does it have, and how many lines of symmetry? 3

16. Give an example (letter or shape) with 3
(i) a line of symmetry but no rotational symmetry (order 1),
(ii) rotational symmetry but no line of symmetry, and explain with T.

17. A point P is 7 units above a horizontal mirror line. Where is its image and how far apart are P and its image? 3

OR

A regular polygon has rotational symmetry of angle 120°. How many sides does it have, and how many lines of symmetry? 3

18. Give an example (letter or shape) with 3
(i) a line of symmetry but no rotational symmetry (order 1),
(ii) rotational symmetry but no line of symmetry, and explain with N.

OR

A point P is 8 units above a horizontal mirror line. Where is its image and how far apart are P and its image? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Study the shape: a regular decagon. Write 5
(a) number of lines of symmetry,
(b) order of rotational symmetry,
(c) smallest angle of rotation,
(d) the angle after 3 such rotations.

OR

Letters of the word OXO: find which letters have vertical symmetry, which have horizontal symmetry and which have rotational symmetry of order 2. Is the word itself symmetric about a vertical mirror? 5

20. Study the shape: a regular 12-gon. Write 5
(a) number of lines of symmetry,
(b) order of rotational symmetry,
(c) smallest angle of rotation,
(d) the angle after 5 such rotations.

OR

Letters of the word HIM: find which letters have vertical symmetry, which have horizontal symmetry and which have rotational symmetry of order 2. Is the word itself symmetric about a vertical mirror? 5

END OF QUESTION PAPER — **Total: 40 Marks**